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C5 REPORT

1. Summary of Research and Code Explanation

In version v0.4, the project requires upgrading the model to solve multivariate and multistep prediction problems. To accomplish this, I researched and implemented the following complex data processing techniques:

Dual Scaler Logic (Multivariate Handling):

Code: Separately initialize `feature_scaler = MinMaxScaler(...)` and `target_scaler = MinMaxScaler(...)`

Explanation: When the data is multivariate (including Open, High, Low, Close, Adj Close, Volume), if only a single shared scaler is used, the input matrix will have a shape of (n, 6). However, the model aims to predict only 1 output feature (Close price). Using a shared scaler will cause a dimensionally error when calling the `inverse_transform` function at the end of the pipeline. I consulted the official Scikit_learn documentation and discussions on StackOverflow to derive the situation of separating the scalers: one for all input variables (Features) and one dedicated exclusively to the target variable (Target).

Multistep Sequence Slicing:

Code: `y_train.append(train_target[i:I + FUTURE_DAYS, 0])`

Explanation: To predict a sequence of closing prices for k days into the future, the target label (`y_train`) can no longer be a single scalar value. By slicing the sequence from i to i + {FUTURE_DAYS}, I transformed the prediction label into a vector containing k consecutive time steps. This structure was referenced from advanced Keras time-series forecasting tutorials.

Output Layer Modification:

Code: `Dense(units=output_steps)` (where `output_steps` is exactly FUTURE_DAYS).

Explanation: Instead of `units=1` as in previous tasks, the final Dense network layer is configured with a number of nodes equal to the number of days k to be predicted. This structural change instructs the Neural Network to simultaneously output multiple future time steps in a single run, rather than iteratively feeding predictions back into the model.

2. Summary of Experiment Results

Below are the observations and evaluations drawn from experimenting with the multivariate and multistep prediction problems:

Evaluating Multivariate vs. Univariate Models:

Adding features such as Trading Volume and price fluctuation margins (High/Low) helps the model analyze much more comprehensively compared to the previous version, which only used the Close price. Experimental results show that the multivariate model reacts faster to sharp price drops if those drops are accompanied by a sudden surge in selling volume.

Evaluating Accuracy across Time Steps (k days):

Short-term (k=3 days): The model closely follows the actual trend, with low latency and a minimal error margin (Loss).

```
Predicted Close prices for the next 3 days:  
[113.571976 111.7524 111.85732 ]  
PS D:\COS30018\COS30018-Group3.4\Source\C5>
```

Medium-term (k=5 days): The model still captures the general direction of the trading week well. However, it begins to struggle with predicting sudden pivot points on the 4th or 5th day.

```
Predicted Close prices for the next 5 days:  
[111.52597 110.582466 112.12764 110.74309 108.752106]  
PS D:\COS30018\COS30018-Group3.4\Source\C5>
```

Long-term (k=10 days): The prediction results tend to converge into a flattened average asymptote (flatline). This indicates that the direct multistep prediction approach significantly loses accuracy if the forecast horizon is too far ahead. In the future, integrating an Attention mechanism could be considered to overcome this weakness.

```
Predicted Close prices for the next 10 days:  
[110.75822 111.290474 110.06199 111.8149 111.42378 110.72685  
110.722206 109.21343 109.22174 108.30683 ]  
PS D:\COS30018\COS30018-Group3.4\Source\C5>
```